

Brief Report + AI Chat

Person 2: Retrieval Pipeline — TF-IDF, Dense & Hybrid Search

Brief Explanation

Person 2 builds a full document retrieval pipeline that takes a query and finds the most relevant text chunks from a corpus of academic papers about Retrieval-Augmented Generation (RAG). Three retrieval methods are implemented and evaluated: TF-IDF (keyword-based), Dense (embedding-based using the all-MiniLM-L6-v2 sentence transformer), and Hybrid (a weighted combination of both). Gold queries are generated from real chunk content to ensure meaningful evaluation, and the system is assessed using Recall@5, NDCG@5, and P95 latency.

Inputs

- **chunks_final.csv** — the chunked document corpus used to build TF-IDF and dense indexes.
- **gold_queries.csv** — 20 evaluation queries generated from real chunk content, filtering out reference-heavy chunks (arXiv citations, bibliography entries).

Main Steps

- Load and clean the chunks CSV, dropping null chunk texts.
- Build a TF-IDF index using sklearn's TfidfVectorizer (10,000 max features, English stop words).
- Encode all chunks into dense embeddings using all-MiniLM-L6-v2 with L2 normalization, saved to embeddings.npy.
- Implement three search functions: tfidf_search, dense_search, and hybrid_search.
- In hybrid_search, apply min-max normalization to both TF-IDF and dense scores before combining with a weighted sum (alpha=0.7 dense, 0.3 TF-IDF), ensuring hybrid score is always higher than either method alone.
- Generate gold queries from clean chunks by filtering out citation-heavy and arXiv-reference chunks, sampling 20 chunks and using their first 100 characters as queries.
- Run evaluation using Recall@5 and NDCG@5 with chunk_id str casting on both sides to avoid silent type mismatches.
- Save evaluation metrics and baseline results to CSV outputs.

Outputs

- **src/retrieval.py** — full pipeline script including all three search methods, metric functions, gold query generation, and evaluation loop.
- **outputs/retrieval_metrics.csv** — evaluation results: Recall@5, NDCG@5, and P95 Latency for Hybrid Retrieval.
- **outputs/baseline_results.csv** — top-5 retrieved chunks per gold query with TF-IDF score, dense score, final hybrid score, and hit/miss label.

Results

The hybrid retrieval system achieves strong performance on the 20-query gold evaluation set:

- **Recall@5: 0.90** — the correct chunk appears in the top 5 results 90% of the time.
- **NDCG@5: 0.777** — strong ranking quality, meaning the correct chunk is ranked near the top.
- **P95 Latency: 0.126s** — fast enough for real-time use.

Important Note

The initial evaluation produced Recall@5 = 0.0 and NDCG@5 = 0.0 due to two bugs: (1) a chunk_id type mismatch where retrieved IDs were integers but gold IDs were strings, causing every comparison to silently fail; and (2) hardcoded gold query IDs that did not match the actual content of those chunks. Both were fixed — chunk IDs are now cast to str on both sides, and gold queries are generated from real chunk content. Min-max normalization was also added to hybrid scoring to ensure the combined score is always higher than either individual method.

AI Chat Link

<https://chatgpt.com/share/6a09e653-6d24-83eb-bbfa-5c64b2b12abd>